



Shore

Year 12
Term II Examination
May 2014

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations
- Start each of questions 11–16 in a new writing booklet
- Write your examination number on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

Exam Number:

Set:

Total marks – 100

Section I Pages 2–5

10 marks

- Attempt questions 1–10
- Allow about 15 minutes for this section

Section II Pages 6–13

90 marks

- Attempt questions 11–16
- Allow about 2 hours and 45 minutes for this section

Note: Any time you have remaining should be spent revising your answers.

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 It is known that for a pair of rational values a and b , $\frac{2}{2+\sqrt{3}} = a + b\sqrt{3}$.

What is the value of a ?

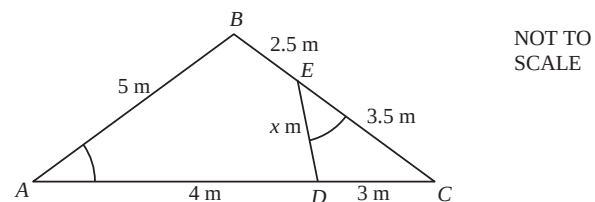
- (A) $a = -4$
- (B) $a = -2$
- (C) $a = 2$
- (D) $a = 4$

- 2 For what value of x does $a^{2x+1} = \frac{1}{a^3}$?

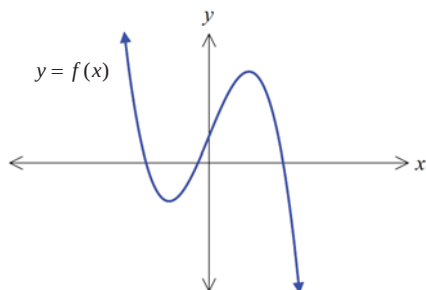
- (A) $x = -2$
- (B) $x = -\frac{1}{2}$
- (C) $x = 1$
- (D) $x = 2$

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- 3 In the diagram below $\angle CAB = \angle DEC$. What is the value of x ?



- (A) 2.5
(B) 2.8
(C) 3.6
(D) 4.375
- 4 Which of the following graphs is a possible gradient function of the function drawn below?



- (A)
- (B)
- (C)
- (D)

- 5 What is the value of $\log_{16} 32$?

- (A) $\frac{1}{2}$
(B) $\frac{4}{5}$
(C) $\frac{5}{4}$
(D) 2

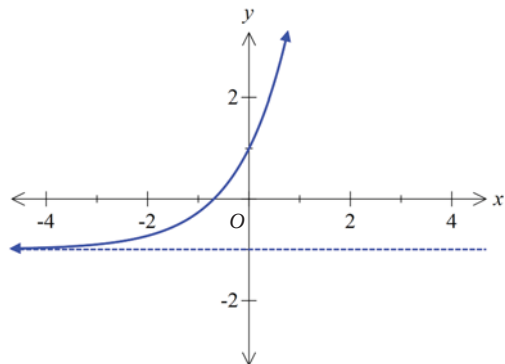
- 6 Which of the following expressions correctly uses Simpson's rule to approximate $\int_1^5 \frac{dx}{x}$ using 5 function values?

- (A) $\frac{1}{3} \left[\left(\frac{1}{1} + \frac{1}{5} \right) + 4 \left(\frac{1}{2} + \frac{1}{4} \right) + 2 \left(\frac{1}{3} \right) \right]$
(B) $\frac{3}{2} \left[\left(\frac{1}{1} + \frac{1}{5} \right) + 4 \left(\frac{1}{2} + \frac{1}{4} \right) + 2 \left(\frac{1}{3} \right) \right]$
(C) $\frac{3}{2} \left[\left(\frac{1}{1} + \frac{1}{5} \right) + 2 \left(\frac{1}{2} + \frac{1}{4} \right) + 4 \left(\frac{1}{3} \right) \right]$
(D) $\frac{1}{3} \left[\left(\frac{1}{1} + \frac{1}{5} \right) + 2 \left(\frac{1}{2} + \frac{1}{4} \right) + 4 \left(\frac{1}{3} \right) \right]$

- 7 What is the primitive of $3x^2 - \frac{1}{x^2}$?

- (A) $x^3 - x^{-1} + C$
(B) $6x + 2x^{-3} + C$
(C) $x^3 + x^{-1} + C$
(D) $x^3 - \ln x + C$

- 8 Which of the following is a possible equation for the graph drawn below?



- (A) $y = e^{2x} - 1$
- (B) $y = 1 - 2e^x$
- (C) $y = 2e^{-x} - 1$
- (D) $y = 2e^x - 1$
- 9 Consider geometric series $1 - 2 + 4 - 8 + 16 - \dots$.
The sum to n terms of this series is 2 796 203. For what value of n is this true?
- (A) 21
- (B) 22
- (C) 23
- (D) no possible value.
- 10 What is the equation of the directrix for the parabola $(x - 2)^2 = -20(y + 1)$?
- (A) $y = -3$
- (B) $y = 4$
- (C) $x = 6$
- (D) $x = 7$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

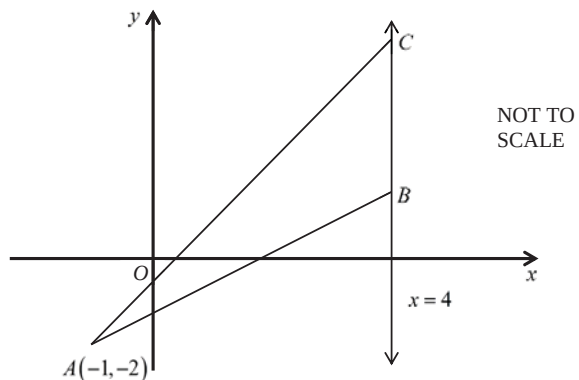
In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Evaluate, correct to 3 significant figures, $\frac{\sqrt{\pi} - 1}{\sin 30^\circ}$. 2
- (b) Factorise $x^2 - 1 - ax - a$. 2
- (c) Find the gradient of the tangent to the curve $y = 4 \ln x$ at the point where $x = e^2$. 2
- (d) Show that $f(x) = \frac{x^2 + 2}{x}$ is an odd function. 2
- (e) Find the equation of the line that passes through the point (1, 2) and is perpendicular to $2x - y + 3 = 0$. 2
- (f) Find the values of x for which the geometric series $(1 - x) + (1 - x)^2 + (1 - x)^3 + \dots$ has a limiting sum? 3
- (g) Find $\int (e^{5x+4} - 2) dx$. 2

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) The points, A , B and C are shown on the number plane below. The point A has coordinates $(-1, -2)$. Points B and C lie on the line $x = 4$.



- (i) The equation of the line AB is given as $3x - 5y - 7 = 0$.
Show that the point B has coordinates $(4, 1)$. 1
- (ii) Given the area of $\triangle ABC$ is 22.5 units^2 , find the length of AC . 3
- (b) The quadratic equation $y = x^2 - \sqrt{2}x - 2$ has the roots α and β .
- (i) Find $\alpha + \beta$. 1
- (ii) Find $\alpha\beta$. 1
- (iii) Find $\alpha^2 + \beta^2$. 2

Question 12 continues on page 8

Question 12 (continued).

- (c) Differentiate $y = \frac{x^2}{e^{3x}}$ with respect to x . Write your answer in simplest form. 3
- (d) (i) Differentiate $y = \sqrt{16 - x^2}$ with respect to x . 2
- (ii) Hence, or otherwise, find $\int \frac{2x}{\sqrt{16 - x^2}} dx$. 2

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) Samantha's parents opened a bank account for her when she was born and deposited \$1 in it. They deposited \$2 on her first birthday, \$4 on her second birthday and so on, with the deposits doubling each birthday. The parents discontinued making deposits after they made a deposit exceeding a million dollars. 3

On what birthday did the deposit exceed a million dollars?

- (b) Find $\int_0^4 x^3 + \frac{1}{\sqrt{x}} dx$. 3

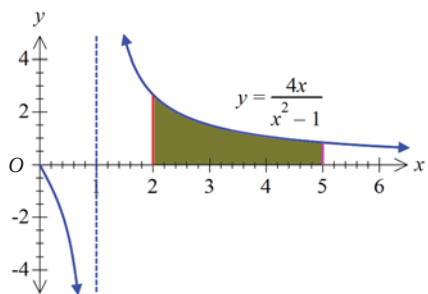
- (c) Evaluate $\lim_{x \rightarrow 3} \frac{3-x}{x^2-9}$. 2

- (d) Everyonetown had a population of 50 000 people on the 1st of January 2009. The population has grown at a constant rate of 9% every year.

- (i) What was the population of Everyonetown on the 1st of January 2014? 2

- (ii) If this population continues to grow at 9% every year, in what year will the population first exceed 100 000 people? 2

- (e) Find the exact value of the area between the curve $y = \frac{4x}{x^2-1}$, the x-axis and the lines $x = 2$ and $x = 5$. 3



Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) Consider the parabola $y^2 - 2y = 12x - 37$. 2

- (i) Rewrite this equation in the form $(y-k)^2 = 4a(x-h)$. 2

- (ii) Hence sketch the graph of the parabola $y^2 - 2y = 12x - 37$, showing the coordinates of the focus. 2

- (b) Consider the function $f(x) = x(x-3)^2$.

- (i) Show that $f'(x) = 3(x-3)(x-1)$. 1

- (ii) Find all stationary points on $y = f(x)$ and determine their nature. 3

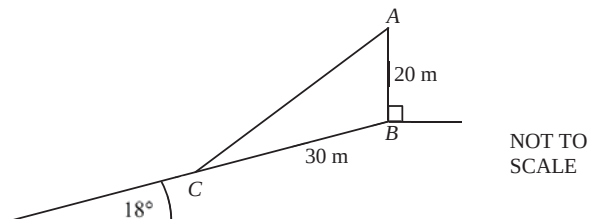
- (iii) Draw a neat sketch of $y = f(x)$, showing all intercepts and stationary points. 2

- (c) A function $f(x)$ has a second derivative $f''(x) = 36x + 6$. Find the equation of $f(x)$ if $y = f(x)$ has a stationary point at $(0, -1)$. 3

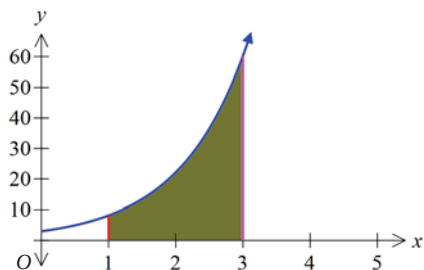
- (d) Solve $|x+1| = |2x-2|$. 2

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) A radio mast AB , of height 20 metres, stands at the top of a slope which is inclined at 18° to the horizontal. The mast is supported by the wire AC which is attached to the point C on the slope. $BC = 30$ metres.

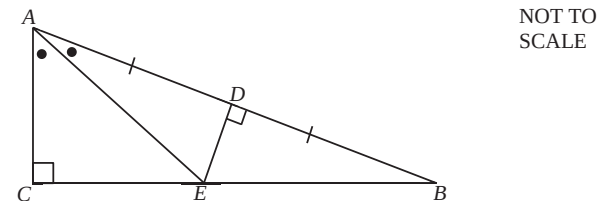


- (i) Calculate the size of $\angle ABC$. 1
- (ii) Find the length of the wire AC . Write your answer correct to the nearest metre. 2
- (b) Show that $\tan^2 x + 1 + \tan x \sec x = \frac{1 + \sin x}{\cos^2 x}$. 2
- (c) The area under the curve $y = 3e^x$ between the values of $x = 1$ and $x = 3$ is rotated about the x -axis. Find the exact volume of the solid of revolution formed. 3



Question 15 continues on page 12

- (d) The diagram below shows a right-angled triangle ABC with $\angle ACB = 90^\circ$. The point D is the midpoint of AB , and E is the point where the perpendicular drawn from D meets BC . AE bisects $\angle DAC$.



Copy or trace the diagram into your booklet.

- (i) Prove that $\triangle ADE \equiv \triangle BDE$. 2
- (ii) Show that $\angle ABC = 30^\circ$. 2
- (iii) Hence find the exact ratio $CB : EB$. 3

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

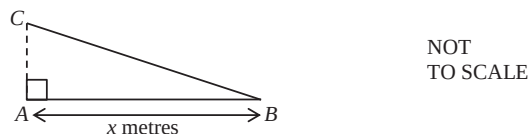
- (a) Stephanie borrowed \$50 000 to start a business. For the first year of the loan Stephanie made no repayments. The loan accrued interest at 9% per annum, compounded monthly. Stephanie then made equal monthly instalments of \$ M for the next 5 years. Let A_n be the amount owing after n months.

(i) Show that the amount owing after 15 months is given by 2

$$A_{15} = 50\,000(1.0075)^{15} - M(1 + 1.0075 + 1.0075^2).$$

(ii) Calculate Stephanie's monthly repayment of \$ M correct to the nearest cent. 3

- (b) A wire of length 10 metres is bent to form the hypotenuse and base of a right angled triangle ABC , as shown in the diagram below. Let the length of the base AB be x metres.



(i) Show that the area of the triangle ABC in square metres is given by 2

$$A = \frac{x}{2} \sqrt{100 - 20x}.$$

(ii) Find the value of x that gives the greatest possible area. 3

- (c) Consider the graphs $y = \sqrt{x}$ and $y = mx$, where m is a constant such that $m > 0$.

(i) Show that the two graphs intersect at $(0, 0)$ and $\left(\frac{1}{m^2}, \frac{1}{m}\right)$. 2

(ii) Hence find the area enclosed by the two graphs in terms of m . 3

End of Paper

1) D

$$\frac{2}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{2(2-\sqrt{3})}{4-3}$$

$$= 4-2\sqrt{3}$$

$$\therefore a=4, b=-2$$

2) A

$$a^{2x+1} = \frac{1}{a^3}$$

$$a^{2x+1} = a^{-3}$$

$$2x+1 = -3$$

$$2x = -4$$

$$x = -2$$

3) A

$$\frac{x}{5} = \frac{3}{6}$$

$$x = 2.5$$

4) B

5) C

$$2^5 = 2^{4x} \quad \log_{16} 32 = x$$

$$5 = 4x \quad \text{or} \quad 32 = 16^x$$

$$x = \frac{5}{4} \quad \ln(32) = x \ln(16)$$

$$x = \frac{\ln(32)}{\ln(16)}$$

$$x = \frac{5}{4}$$

6) A

7) C

$$f(x) = 3x^2 - \frac{1}{x^2}$$

$$= 3x^2 - x^{-2}$$

$$\int f(x) dx = \int 3x^2 - x^{-2} dx$$

$$= x^3 + x^{-1} + C$$

d)

$$f(x) = \frac{x^2+2}{x}$$

$$f(-x) = \frac{(-x)^2+2}{-x}$$

$$= -\frac{x^2+2}{x}$$

$$= -f(x)$$

e)

$$2x - y + 3 = 0$$

$$y = 2x + 3$$

$$m_1 = 2$$

$$m_2 = -\frac{1}{2}, \text{ through } (1, 2)$$

$$y - 2 = -\frac{1}{2}(x - 1)$$

$$2y - 4 = -x + 1$$

$$x + 2y - 5 = 0$$

f)

$$(1-x) + (1-x)^2 + (1-x)^3 + \dots$$

$$r = (1-x)$$

for limiting sum

$$-1 < r < 1$$

$$-1 < 1-x < 1$$

$$-2 < -x < 0$$

$$0 < x < 2$$

g)

$$\int e^{(5x+4)} - 2 dx = \frac{1}{5} e^{(5x+4)} - 2x + C$$

8) D

9) C

$$1-2+4-8+16-\dots$$

$$a=1, r=-2, S_n=2796203$$

$$S_n = \frac{a(r^n-1)}{r-1}$$

$$2796203 = \frac{1((-2)^n-1)}{-2-1}$$

$$-8388609 = (-2)^n - 1$$

$$-8388608 = (-2)^n$$

$$(-1)8388608 = (-1)^n(2)^n$$

$$n = \frac{\ln(8388608)}{\ln(2)} \text{ for } n \text{ odd}$$

$$= 23$$

10) B

11)

$$a) \frac{\sqrt{\pi}-1}{\sin 30^\circ} = 1.544907702\dots$$

$$\approx 1.54$$

$$b) x^2 - 1 - ax - a = (x-1)(x+1) - a(x+1)$$

$$= (x+1)(x-1-a)$$

c) $y = 4 \ln x$

$$\frac{dy}{dx} = \frac{4}{x}$$

$$\text{at } x = e^2$$

$$\frac{dy}{dx} = \frac{4}{e^2}$$

12)

a)

$$i) 3x - 5y - 7 = 0$$

$$\text{at } x = 4$$

$$3(4) - 5y - 7 = 0$$

$$12 - 7 = 5y$$

$$y = 1$$

ii) Area of $\triangle ABC = 22.5$

$$22.5 = \frac{1}{2}(5)BC$$

$$BC = 9$$

$$AC = \sqrt{5^2 + (9+3)^2}$$

$$= 13 \text{ units}$$

b)

$$y = x^2 - \sqrt{2}x - 2$$

$$i) \alpha + \beta = \frac{-b}{a}$$

$$= \frac{-(-\sqrt{2})}{1}$$

$$= \sqrt{2}$$

$$ii) \alpha\beta = \frac{c}{a}$$

$$= \frac{-2}{1}$$

$$= -2$$

$$iii) \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= (\sqrt{2})^2 - 2(-2)$$

$$= 2 + 4$$

$$= 6$$

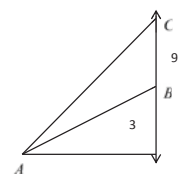
c)

$$y = \frac{x^2}{e^{3x}}$$

$$\frac{dy}{dx} = \frac{e^{3x} 2x - x^2 3e^{3x}}{e^{6x}}$$

$$= \frac{xe^{3x}(2-3x)}{e^{6x}}$$

$$= \frac{x(2-3x)}{e^{3x}}$$



d)

i) $y = \sqrt{16-x^2}$

$$= (16-x^2)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2} \times (16-x^2)^{-\frac{1}{2}} \times -2x$$

$$= \frac{-x}{\sqrt{16-x^2}}$$

ii) $\int \frac{2x}{\sqrt{16-x^2}} dx$

$$= -2 \int \frac{-x}{\sqrt{16-x^2}} dx$$

$$= -2\sqrt{16-x^2} + C$$

13)

a) $T_n = ar^{n-1}$

$$1000000 = 1 \times 2^{n-1}$$

$$1000000 = \frac{2^n}{2}$$

$$2000000 = 2^n$$

$$n = \frac{\ln(2000000)}{\ln(2)}$$

$$= 20.931...$$

$\therefore$ last desposit on her 20th Birthday

b)

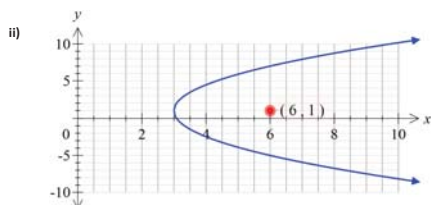
$$\int_0^4 x^3 + \frac{1}{\sqrt{x}} dx$$

$$= \left[\frac{x^4}{4} + 2x^{\frac{1}{2}} \right]_0^4$$

$$= \left[\frac{(4)^4}{4} + 2(4)^{\frac{1}{2}} \right] - \left[\frac{(0)^4}{4} + 2(0)^{\frac{1}{2}} \right]$$

$$= (64 + 4) - 0$$

$$= 68$$



b)

i) $f(x) = x(x-3)^2$

$$f'(x) = 2x(x-3) + (x-3)^2$$

$$= (x-3)[2x+x-3]$$

$$= (x-3)(3x-3)$$

$$= 3(x-3)(x-1)$$

ii)

$$f'(x) = 3(x-3)(x-1)$$

$$= 3x^2 - 12x + 9$$

Stationary points at $f'(x) = 0$

$$x = 3 \text{ and } x = 1$$

$$f(1) = 1(1-3)^2$$

$$= 4$$

$$f(3) = 3(3-3)^2$$

$$= 0$$

$$f''(x) = 6x^2 - 12$$

$$f''(1) = 6(1)^2 - 12$$

$$= -6$$

$$< 0$$

$\therefore$ Maximum Turning point at (1, 4)

$$f''(x) = 6x^2 - 12$$

$$f''(3) = 6(3)^2 - 12$$

$$= 42$$

$$> 0$$

$\therefore$ Minimum Turning point at (3, 0)

c)

$$\lim_{x \rightarrow 3} \frac{3-x}{x^2-9} = \lim_{x \rightarrow 3} \frac{-(x-3)}{(x+3)(x-3)}$$

$$= \lim_{x \rightarrow 3} \frac{-1}{(x+3)}$$

$$= \frac{-1}{(3+3)}$$

$$= -\frac{1}{6}$$

d)

i) $P = 50000(1+0.09)^5$

$$= 76931.19775$$

$$\approx 76931$$

ii) $100000 = 50000(1.09)^n$

$$n = \frac{\ln(2)}{\ln(1.09)}$$

$$= 8.043231...$$

The population will first exceed 100 000 in 2017

e)

$$A = \int_2^5 \frac{4x}{x^2-1} dx$$

$$= 2 \int_2^5 \frac{2x}{x^2-1} dx$$

$$= 2 \left[\ln(x^2-1) \right]_2^5$$

$$= 2 \left[\ln(24) - \ln(3) \right]$$

$$= 2 \ln 8$$

$$= 6 \ln 2 \text{ units}^2$$

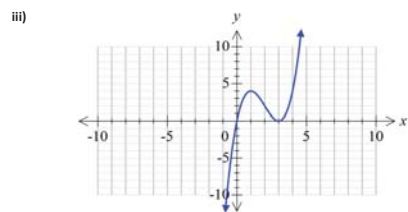
14)

a) $y^2 - 2y = 12x - 37$

i) $y^2 - 2y + 1 = 12x - 37 + 1$

$$(y-1)^2 = 12x - 36$$

$$(y-1)^2 = 12(x-3)$$



c)

$$f''(x) = 36x + 6$$

$$f'(x) = \int f''(x) dx$$

$$= 18x^2 + 6x + C_1$$

$$f'(0) = 0$$

$$0 = 18(0)^2 + 6(0) + C_1$$

$$C_1 = 0$$

$$f'(x) = 18x^2 + 6x$$

$$f(x) = \int f'(x) dx$$

$$= 6x^3 + 3x^2 + C_2$$

$$f(0) = -1$$

$$-1 = 6(0)^3 + 3(0)^2 + C_2$$

$$C_2 = -1$$

$$\therefore f(x) = 6x^3 + 3x^2 - 1$$

d)

$$|x+1| = |2x-2|$$

case 1

$$x+1 = 2x-2$$

$$x = 3$$

case 2

$$x+1 = -2x+2$$

$$x = \frac{1}{3}$$

test $x = \frac{1}{3}$

$$\therefore x = 3, x = \frac{1}{3}$$

15)

a)

$$\begin{aligned} \text{i) } \angle ABC &= 90^\circ + 18^\circ \\ &= 108^\circ \end{aligned}$$

$$\begin{aligned} \text{ii) } AC^2 &= (30)^2 + (20)^2 - 2 \times 30 \times 20 \cos 108^\circ \\ &= 1670.820393... \\ AC &= 40.87566... \\ &\approx 41 \text{ m} \end{aligned}$$

b)

$$\begin{aligned} \tan^2 x + 1 + \tan x \sec x &= \frac{1 + \sin x}{\cos^2 x} \\ LHS &= \tan^2 x + 1 + \tan x \sec x \\ &= \frac{\sin^2 x}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x} + \frac{\sin x}{\cos x} \times \frac{1}{\cos x} \\ &= \frac{1 + \sin x}{\cos^2 x} \\ &= RHS \end{aligned}$$

c)

$$\begin{aligned} y &= 3e^x \\ y^2 &= (3e^x)^2 \\ &= 9e^{2x} \\ V &= \pi \int_1^3 9e^{2x} dx \\ &= \pi \left[\frac{9e^{2x}}{2} \right]_1^3 \\ &= \pi \left[\frac{9e^{2(3)}}{2} - \frac{9e^{2(1)}}{2} \right] \\ &= \frac{9\pi(e^6 - e^2)}{2} \text{ units}^2 \\ &= \frac{9\pi e^2(e^4 - 1)}{2} \text{ units}^2 \end{aligned}$$

d)

$$\begin{aligned} \text{i) } \triangle ADE &\equiv \triangle BDE \\ AD &= BD \text{ (Given)} \\ DE &= DE \text{ (Common)} \\ \angle ADE &= \angle BDE = 90^\circ \\ \therefore \triangle ADE &\equiv \triangle BDE \text{ (SAS)} \end{aligned}$$

$$\text{ii) } \angle CAE = \angle EAD \text{ (AE bisects } \angle BAC)$$

$$\text{let } \angle CAE = \alpha$$

$$\angle EAD = \angle EBD \text{ (matching angles in congruent triangles)}$$

$$3\alpha = 90^\circ \text{ (Angle sum of a triangle ABC)}$$

$$\alpha = 30^\circ$$

$$\angle ABC = 30^\circ$$

iii)

$$\text{let } AB = 2$$

$$\sin 30^\circ = \frac{AC}{2}$$

$$AC = 1$$

$$CB = \sqrt{2^2 - 1^2}$$

$$= \sqrt{3}$$

$$DB = 1$$

$$\cos 30^\circ = \frac{1}{EB}$$

$$EB = \frac{2}{\sqrt{3}}$$

$$CB : EB$$

$$\sqrt{3} : \frac{2}{\sqrt{3}}$$

$$3 : 2$$

16)

a)

$$\begin{aligned} \text{i) } A_1 &= 50000(1.0075)^1 \\ A_2 &= 50000(1.0075)^2 \\ A_3 &= 50000(1.0075)^3 - M \\ A_4 &= [50000(1.0075)^3 - M](1.0075) - M \\ A_4 &= 50000(1.0075)^4 - M(1 + 1.0075) \\ A_5 &= 50000(1.0075)^5 - M(1 + 1.0075 + 1.0075^2) \end{aligned}$$

ii)

$$\begin{aligned} A_2 &= 50000(1.0075)^2 - M(1 + 1.0075 + \dots + 1.0075^{59}) \\ A_2 &= 0 \\ M &= \frac{50000(1.0075)^2}{1 + 1.0075 + \dots + 1.0075^{59}} \\ 1 + 1.0075 + \dots + 1.0075^{59} &= \frac{1(1.0075^{60} - 1)}{0.0075} \\ M &= \frac{50000(1.0075^2)}{1.0075^{60} - 1} \times 0.0075 \\ M &= \frac{375(1.0075^2)}{1.0075^{60} - 1} \\ &= \$1135.28 \end{aligned}$$

b)

$$\begin{aligned} \text{i) } AC &= \sqrt{(10-x)^2 - x^2} \\ &= \sqrt{100 - 20x + x^2 - x^2} \\ &= \sqrt{100 - 20x} \end{aligned}$$

$$\begin{aligned} \text{Area } \triangle ABC &= \frac{1}{2} \times \sqrt{100 - 20x} \\ &= \frac{x}{2} \sqrt{100 - 20x} \end{aligned}$$

ii)

$$\begin{aligned} A &= \frac{x}{2} \sqrt{100 - 20x} \\ &= \frac{x(100 - 20x)^{\frac{1}{2}}}{2} \end{aligned}$$

$$\begin{aligned} \frac{dA}{dx} &= \frac{-10x}{\sqrt{100 - 20x}} + \frac{\sqrt{100 - 20x}}{2} \\ &= \frac{-10x + (100 - 20x)}{2\sqrt{100 - 20x}} \end{aligned}$$

$$\text{Stat points at } \frac{dA}{dx} = 0$$

$$0 = \frac{-10x + (100 - 20x)}{2\sqrt{100 - 20x}}$$

$$0 = 100 - 30x$$

$$x = \frac{10}{3} \text{ m}$$

x	3	$\frac{10}{3}$	4
$\frac{dA}{dx}$	$\frac{-10(3) + (100 - 20(3))}{2\sqrt{100 - 20(3)}}$ $= \frac{-30 + 40}{2\sqrt{40}}$ > 0	0	$\frac{-10(4) + (100 - 20(4))}{2\sqrt{100 - 20(4)}}$ $= \frac{-40 + 20}{2\sqrt{40}}$ < 0
			

$$\text{Therefore max area when } x = \frac{10}{3} \text{ m.}$$

c)

i)

$$y = \sqrt{x}$$

$$y = mx$$

$$\sqrt{x} = mx$$

$$0 = \sqrt{x} - mx$$

$$= x \left(x^{-\frac{1}{2}} - m \right)$$

$$x = 0$$

$$\therefore y = 0$$

and

$$x^{-\frac{1}{2}} = m$$

$$x = m^{-2}$$

$$= \frac{1}{m^2}$$

$$\therefore y = \sqrt{\frac{1}{m^2}}$$

$$= \frac{1}{m}$$

ii) $\therefore$ The graphs intersect at $(0,0)$ and $\left(\frac{1}{m^2}, \frac{1}{m}\right)$.

$$A = \int_0^{\frac{1}{m^2}} (\sqrt{x} - mx) dx$$

$$= \left[\frac{2x^{\frac{3}{2}}}{3} - \frac{mx^2}{2} \right]_0^{\frac{1}{m^2}}$$

$$= \left[\frac{2 \left(\frac{1}{m^2} \right)^{\frac{3}{2}}}{3} - \frac{m \left(\frac{1}{m^2} \right)^2}{2} \right] - 0$$

$$= \frac{2 \left(\frac{1}{m^3} \right)}{3} - \frac{\left(\frac{1}{m^3} \right)}{2}$$

$$= \left(\frac{2}{3m^3} \right) - \left(\frac{1}{2m^3} \right)$$

$$= \frac{1}{6m^3} \text{ units}^2$$